

Cardiovascular Health Across the Life Course for Individuals With Breast Cancer

A Review

Rina Mauricio, MD; Anita Deswal, MD, MPH, MBBS; Kathryn J. Ruddy, MD, MPH; Nausheen Akhter, MD; Ming Hui Chen, MD, MMSc; Imo A. Ebong, MD, MS; Niti R. Aggarwal, MD; Colleen M. Harrington, MD; Carlos H. Barcenas, MD, MSc; Melissa Pulfer Mitchell, MD, PhD; Leslie Cho, MD; Anju Nohria, MD, MSc

IMPORTANCE Breast cancer is the most common cancer diagnosed in women. Therapies for breast cancer are evolving rapidly, and several are associated with cardiovascular adverse effects, can exacerbate cardiovascular risk factors, or can increase the risk of cardiovascular disease.

OBSERVATIONS Advances in screening and treatment have led to earlier diagnosis and improved disease-free survival. For long-term survivors of breast cancer, cardiovascular adverse effects can pose a greater threat to overall long-term health than the cancer itself. Additionally, the increasing incidence of breast cancer in younger, premenopausal women has created a distinct demographic with unique medical needs.

CONCLUSIONS AND RELEVANCE Cardiology practitioners need to remain updated on rapidly evolving breast cancer therapies to care for this growing, unique patient population. The field of cardio-oncology has evolved to address the cardiovascular needs of patients and survivors.

JAMA Cardiol. 2026;11(9):867-879. doi:10.1001/jamacardio.2026.2546
Published online July 29, 2026.

← Viewpoint page 795

← JAMA Cardiology Patient
Page page 901

+ CME at jamacmelookup.com

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: Rina Mauricio, MD, Department of Medicine, Division of Cardiology, University of Texas Southwestern Medical Center, 2001 Inwood Rd, Dallas, TX 75390 (rina.mauricio@utsouthwestern.edu).

Breast cancer (BC) is the most common cancer diagnosed in women,^{1,2} constituting 15.5% of all new cancer cases and 6.9% of all cancer deaths in the US in 2024.³ Although BC is more common in postmenopausal women (69%), approximately one-third of BC cases occur in premenopausal women.⁴ In the US, female BC incidence has increased by 1.1% per year in women younger than 50 years and by 0.5% per year in women aged 50 years and older.¹ Concomitantly, 5-year survival rates have increased from 75% in the mid-1970s to 91% in 2024, and survivors range in age from older than 65 years (estimated two-thirds of survivors, approximately 2.7 million women) to younger than 50 years (6% of survivors).^{2,3}

Treatment for BC has evolved considerably over time (Figure 1), and BC therapies have varying short- and long-term cardiovascular (CV) adverse effects⁵⁻⁴³ (Table 1). CV toxicity is often considered in the context of cancer therapy-related cardiac dysfunction (CTRCD), which can range from asymptomatic abnormalities in cardiac biomarkers and function to overt heart failure (HF).⁴⁴ However, other toxic effects, including QTc prolongation, hypertension, cardiometabolic abnormalities, and myocarditis, also occur.

Emerging evidence suggests shared risk factors, such as diabetes, obesity, tobacco and alcohol use, and pathophysiology between CV disease (CVD) and BC.⁴⁵ Preexisting CV health influences cancer treatment decisions and affects cancer outcomes.^{5,46} As the number of survivors of BC grows, CVD prevention and management are becoming increasingly important.⁵

This review is intended for cardiologists treating individuals at risk for both CVD and BC, patients undergoing active treatment for

BC, and survivors of BC who are at risk for or living with CVD, partly related to BC therapies.

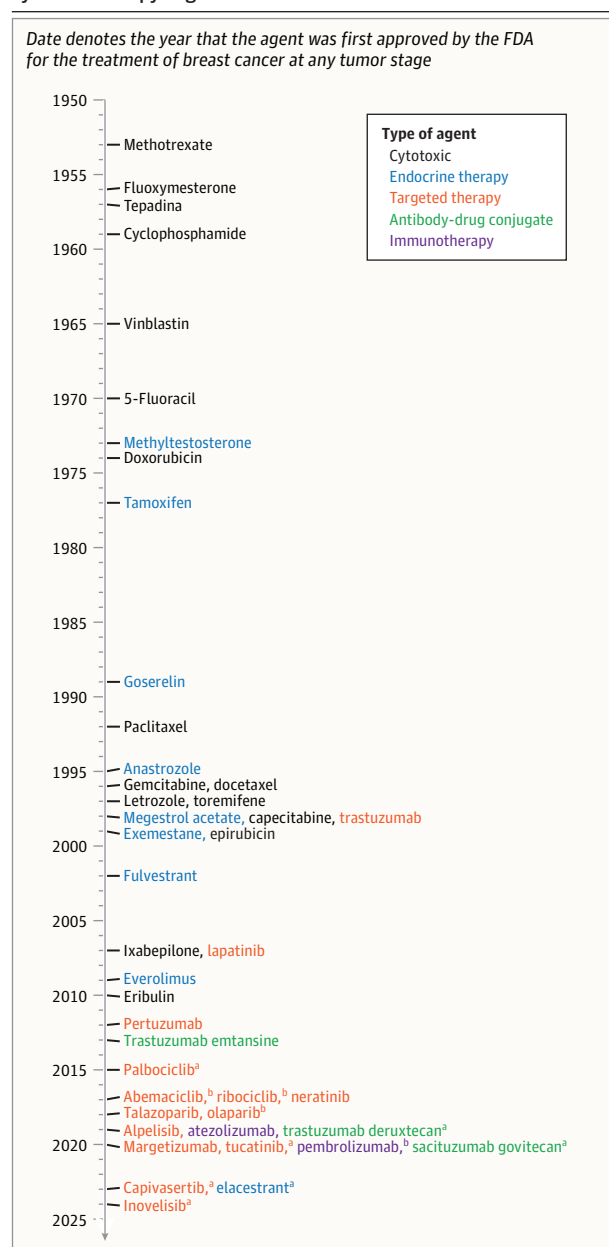
Preexisting CVD and Risk of Malignancy: Reverse Cardio-Oncology

Poor CV health has been associated with a higher risk of cancer (reverse cardio-oncology), suggesting shared pathways.⁴⁵⁻⁴⁸ Large epidemiologic studies have consistently demonstrated a higher incidence of malignancy among individuals with known CVD.⁴⁹ Premature CVD has been associated with an almost 4-fold increased risk of BC,⁵⁰ and incident CVD has been associated with an increased risk of recurrence and worse overall mortality in patients with BC.⁵¹ Preclinical models support these findings. Mouse models of left ventricular (LV) dysfunction and myocardial infarction have shown increased rates of malignancy.^{51,52} Cellular models have identified post-myocardial infarction alterations in cell pathways, prompting tumorigenesis.⁵³⁻⁵⁵ Better understanding of pathways between CVD and BC could improve outcomes for both diseases.

Overview of BC Treatment

Since 2020, more than 10 new BC therapies have been approved by the US Food and Drug Administration (Figure 1). Early-stage (stage 1-3) BC is generally treated with curative intent using a combina-

Figure 1. Timeline of US Food and Drug Administration (FDA)-Approved Systemic Therapy Regimens for Breast Cancer



This figure outlines FDA-approved systemic therapy regimens for breast cancer. The date denotes the year in which the agent was first approved by the FDA for the treatment of breast cancer at any tumor stage. Some listed medications later received FDA approval (dates not shown) for different indications, such as early breast cancer, combination regimens, or different presentations (ie, nab-paclitaxel).

^aOnly approved for metastatic disease.

^bApproved for early-stage disease.

tion of surgery, systemic therapy, and, for more than 50% of patients, radiation therapy. Metastatic BC (stage 4, or distantly recurrent) is almost always incurable and is managed with systemic therapies to slow progression and relieve symptoms. Disease progression is inevitable with metastatic BC, often requiring multiple treatment changes.

BC tumor subtype is critically important for therapy selection. For estrogen receptor (ER)-positive and/or progesterone receptor (PR)-positive tumors, chemotherapy is not always warranted because endocrine therapy (ET) with or without targeted therapy may be adequate. For ERBB2 (formerly HER2)-positive cancers, ERBB2-targeted therapy is used in combination with chemotherapy in both early-stage and metastatic settings. Triple-negative cancers (ER, PR, and ERBB2 negative) are typically treated with systemic chemotherapy. Since 2021, pembrolizumab, an immune checkpoint inhibitor (ICI), has been used with taxane or platinum-based chemotherapy for metastatic BC that expresses programmed death-ligand 1.

Chemotherapy for early-stage breast cancer is often given in the neoadjuvant (preoperative) setting, especially for ER-negative and/or ERBB2-positive disease. For disease that is triple negative (ER, PR, and ERBB2 negative), this usually includes an anthracycline. Adjuvant (postoperative) chemotherapy can be administered instead of or in addition to neoadjuvant chemotherapy but is less likely to include anthracyclines. Since 2021, pembrolizumab is often added to neoadjuvant anthracycline-based chemotherapy for triple-negative BC to improve pathologic complete response rates.

Review of Cardiotoxic BC Treatments

Table 1 outlines CV toxic effects associated with commonly used BC therapies. Select treatments are discussed below.

Anthracyclines

Anthracycline use has declined in the US but remains common elsewhere due to limited access to newer and costlier drugs.⁶ In the nonmetastatic setting, anthracyclines are usually prescribed at a cumulative doxorubicin-equivalent dose of 240 mg/m², which is less frequently associated with CTRCD. In metastatic BC, anthracyclines are typically used as single agents later in the disease course and cumulative doses may be higher (eg, 480 mg/m² of doxorubicin), with higher rates of CTRCD. The mechanisms leading to anthracycline cardiotoxicity are reviewed elsewhere,^{56,57} with risk factors for anthracycline-induced CTRCD listed in Table 1. Anthracyclines cause dose-dependent CTRCD, which presents most commonly within the first year after completion of therapy or, less commonly, many years later.⁴⁴ Studies have shown that partial, and less frequently complete, LV ejection fraction (LVEF) recovery can occur, especially if CTRCD is detected early and treated with guideline-directed medical therapy (GDMT) for HF.^{7,57}

ERBB2-Targeted Therapy

Trastuzumab, a humanized monoclonal antibody, reduces the risk of recurrence in early-stage ERBB2-positive BC and more than doubles the average overall survival in metastatic ERBB2-positive BC.⁵⁸ The initial trial of trastuzumab, administered concurrently with anthracyclines, in metastatic BC showed high rates of cardiac dysfunction (28%) and New York Heart Association class III or IV HF (19%).⁸ Consequently, trastuzumab and other ERBB2-targeted agents are no longer given concurrently but are used sequentially if anthracyclines are required, with significantly lower risk of CTRCD. Risk factors for ERBB2 therapy-associated CTRCD are outlined in Table 1.^{9,59}

While nearly all patients with ERBB2-positive BC receive trastuzumab, some receive additional ERBB2-targeted therapies

Table 1. Cardiovascular (CV) Effects of Breast Cancer Therapies

Class of therapy	CV effects	Timing of CV toxic effects	Risk factors for development of CV toxic effects and other comments
Anthracyclines			
Doxorubicin, epirubicin	Dose-dependent CTRCD; incidence varies based on definition of CTRCD used; HF risk with modern regimens (240 mg/m ² of doxorubicin equivalent) is <2% at 5 y with approximate 10% incidence of all CTRCD (symptomatic and asymptomatic) ⁶ ; less commonly, arrhythmias and QT interval prolongation ¹⁸	Concurrent and delayed clinical presentation (majority within 1 y of treatment completion; may occur later, especially with higher doses) ⁷	CTRCD risk higher when trastuzumab is also used; other risk factors for CTRCD include cumulative dose, thoracic RT, age extremes (younger and older), hypertension, preexisting heart disease, including a low normal LVEF; often avoided if LVEF <45%-50%, especially if symptomatic with HF
Antimetabolites			
Taxanes	QT interval prolongation, bradycardia, atrial fibrillation	Concurrent	
Fluoropyrimidines	Coronary vasospasm, up to 10% ¹⁹ ; risk dependent on route of delivery, usually seen with initial exposure but may persist after treatment cessation; myocarditis, arrhythmia, and peripheral arterial toxic effects are rare ²⁰ ; hypertension	Concurrent, but can persist after treatment cessation	
Alkylating agents			
Cyclophosphamide	Uncommon cardiac effects of HF, myocarditis, or atrial fibrillation	Concurrent	
Cisplatin	QT prolongation	Concurrent	
ERBB2-targeted therapy			
Trastuzumab	Overall CTRCD of 4%-14% and symptomatic HF 0.8%-4% in clinical trials and contemporary cohorts ^{9,17,21}	Concurrent or early after completion	Rates of CTRCD higher with concurrent anthracycline use ⁸ ; now used sequentially; factors that increase risk for CTRCD ^{15,16}
Pertuzumab	Addition of pertuzumab to trastuzumab does not significantly increase risk of CTRCD ^{22,23}		include anthracycline use, older age, borderline or low pretreatment LVEF, CV risk factors or established CVD, moderate or severe valvular heart disease, cigarette smoking, obesity; can prolong QT interval with long-term treatment ¹⁸
Trastuzumab-emtansine, trastuzumab-deruxtecan	CTRCD risk 1%-2% in lower-risk patients in clinical trials ²³		
ET			
SERMs			
Tamoxifen	Increased VTE risk (2-3 fold) ²⁴⁻²⁶ ; potential reduction in coronary ischemic events	Concurrent	Conflicting data on modest increase in stroke likely counterbalanced by modest benefit on coronary ischemic events ²⁴ ; LDL decreased, apolipoprotein A-1 decreased, TGs increased ³² ; rarely QT interval prolongation can occur, usually with other agents or conditions causing the same ¹⁸
Als			
Letrozole, anastrozole, exemestane	In meta-analyses of RCTs, modest increase in CV events ⁹ (relative 16%-19% increase) and ischemic heart disease (relative 30% increase) with Als compared with tamoxifen; however, no significant increase in CV events compared with placebo ^{27,28}	Concurrent	Possible increase in lipid levels with Als compared with tamoxifen, which may also differ between Als, but data are conflicting ^{11,28,33-37} ; increase in CV events compared with tamoxifen, more prominent with anastrozole
Ovarian suppression			
GnRH agonist, oophorectomy	Potential CV risks of hyperlipidemia, CAD, hypertension, and diabetes in premenopausal women ²⁹⁻³¹	Concurrent, potentially long-term after oophorectomy	Premature menopause increases the risk of developing CV risk factors or atherosclerotic CVD or HF; recognized as a CVD risk-enhancing factor ^{12,13,38}
SERDs			
Fulvestrant, elacestrant	No significant CV risks noted		
Cyclin-dependent kinase 4/6 inhibitor			
Ribociclib	Concentration-dependent QTc interval prolongation reported, 3%-6% with ribociclib ¹⁰	Concurrent	QTc interval monitoring is recommended for ribociclib
Palbociclib, abemaciclib	Increase in VTE and potentially arterial thrombosis ³⁹	Concurrent	RCTs have not suggested other significant CV AEs with these agents; observational data suggest there may be more CV events ⁴⁰
Tyrosine kinase inhibitors (ERBB2 [formerly HER2] targeted)			
Tucatinib	No significant CV risk ^{41,42}		
Neratinib, lapatinib	Possible modest risk of decreased LVEF or increased QTc interval ⁴²	Concurrent	CTRCD more common if used with trastuzumab
mTOR inhibitor			
Everolimus	Metabolic syndrome, hyperglycemia, dyslipidemia, hypertension (>30%) ^{10,23}	Concurrent	Used in combination with ET

(continued)

Table 1. Cardiovascular (CV) Effects of Breast Cancer Therapies (continued)

Class of therapy	CV effects	Timing of CV toxic effects	Risk factors for development of CV toxic effects and other comments
AKT inhibitor			
Capivasertib	Hyperglycemia (grade 3-4 in approximately 3%)	Concurrent	Used in combination with ET
Phosphatidylinositol 3-kinase inhibitor			
Alpelisib	Hyperglycemia (grade 3-4 in approximately 35%) ⁴³	Concurrent	Used in combination with ET
Inavolisib	Hyperglycemia (grade 3-4 in approximately 6%)		
Thoracic RT	Increased risk of coronary events Incidence may be lower with changes in RT techniques and dosing over the years	Intermediate to delayed	Risk factors for cardiotoxic effects: cumulative or mean dose received to the heart, anthracycline use, age (younger), preexisting CV risk factors
ICIs			
Pembrolizumab	Myocarditis in <1% with single ICI; pericarditis; potential long-term accelerated atherosclerosis ¹⁴	Myocarditis and pericarditis are concurrent; usually occur early after initiation of ICI (median, approximately 30 d), but can be later during therapy	
PARP inhibitors	No cardiotoxic signal noted		

Abbreviations: AEs, adverse events; AIs, aromatase inhibitors; AKT, protein kinase B; CAD, coronary artery disease; CTRCD, cancer therapy-related cardiac dysfunction; CVD, cardiovascular disease; ET, endocrine therapy; GnRH, gonadotropin-releasing hormone; HF, heart failure; ICIs, immune checkpoint inhibitors; LDL, low-density lipoprotein; LVEF, left ventricular ejection fraction; mTOR, mechanistic target of rapamycin;

PARP, poly(adenosine diphosphate-ribose) polymerase; RCTs, randomized clinical trials; RT, radiation therapy; SERDs, selective estrogen receptor degraders; SERMs, selective estrogen receptor modulators; TGs, triglycerides; VTE, venous thromboembolism.

^a Varying CV events included in different trials and observational studies.

either in combination with trastuzumab (eg, pertuzumab or tucatinib) or after progression while receiving trastuzumab (eg, trastuzumab deruxtecan, trastuzumab emtansine, or neratinib). These newer agents do not appear to increase the risk of CTRCD when used with trastuzumab and have lower reported rates of CTRCD (1%-2%) relative to trastuzumab.⁶⁰ ERBB2 therapy-associated CTRCD typically occurs during treatment, is often asymptomatic, and is frequently, but not always, reversible with initiation of GDMT for HF (especially if anthracyclines have not been used) and/or after interrupting ERBB2 therapy if appropriate.⁶¹ Permissive cardiotoxicity has been primarily studied in the context of trastuzumab-associated CTRCD and is discussed below.⁶²⁻⁶⁵

ET

ET has proven efficacy in patients with ER-positive BC and is used for prolonged periods of 5 to 10 years.⁶⁶ ET either interferes with ER signaling (selective ER modulators [SERMs] or selective ER degraders [SERDs]) or prevents the production of estrogen (aromatase inhibitors [AIs]).^{10,11} ET options for premenopausal patients with early-stage BC include an AI with ovarian function suppression or tamoxifen (SERM) with or without ovarian function suppression. For postmenopausal patients, AIs are the preferred adjuvant ET.⁶⁶ SERDs (eg, fulvestrant and elacestrant) and SERMs are typically used after AI treatment failure.

ET has been associated with adverse changes in blood pressure, lipid levels, weight, adiposity, and body composition.^{11,67,68} ET, especially tamoxifen, may increase the risk of type 2 diabetes, partly due to weight gain. However, obesity, a risk factor for both BC and type 2 diabetes, may confound this association. Data regarding AI-associated CVD risk is mixed. Some studies have suggested increased risk,⁶⁹ particularly after long-term use (>4 years),⁷⁰ while others have shown no increased risk of clinical or subclinical atherosclerotic CVD.^{71,72} Tamoxifen has been associated with a reduced risk of CVD; thus, the higher risk seen with AIs compared with

tamoxifen may reflect the protective effect of tamoxifen rather than harm from AIs.¹⁰ Although data on CV risk with AIs are inconclusive, the 2022 European Society of Cardiology (ESC) guidelines recommend annual monitoring of blood pressure and lipid levels to reduce CV risk in these patients.⁵⁹

Ovarian function suppression can be achieved with medications, bilateral oophorectomy, or bilateral ovarian radiation. Several trials have demonstrated that the addition of ovarian function suppression to ET improves disease-free and overall survival in premenopausal patients with ER-positive and PR-positive BC, with benefits seen at extended follow-up.⁷³⁻⁷⁵ Bilateral oophorectomy or ovarian radiation results in permanent premature menopause in premenopausal patients. Natural and surgical premature menopause increases the risk of future CVD^{12,76,77} and is thus a risk enhancer for CVD in recent guidelines.¹³ The Cardiac Outcomes With Near-Complete Estrogen Deprivation (CROWN) study is an ongoing multicenter prospective cohort study evaluating CV toxic effects with AI use and ovarian function suppression in premenopausal women with hormone receptor-positive BC.⁷⁸

Targeted Therapies

Cyclin-dependent kinase 4/6 inhibitors (eg, ribociclib and abemaciclib) are approved as adjuvant therapy in combination with ET for high-risk, early-stage BC. In particular, ribociclib has a 3% to 6% increased risk of QTc interval prolongation, thus requiring QTc interval monitoring prior to and 2 weeks after initiation of this medication and with upward dose adjustments and/or when other QTc interval-prolonging drugs are added.¹⁰ For patients with metastatic disease, other targeted therapies, including phosphatidylinositol 3-kinase inhibitors (eg, alpelisib and inavolisib), AKT inhibitors (capivasertib), and mechanistic target of rapamycin (mTOR) inhibitors (everolimus) are combined with ET based on tumor genomics. Additional details on known CV toxic effects of these agents are listed in Table 1.

Immunotherapy

Following the landmark KEYNOTE-522 trial, ICIs in combination with chemotherapy were recently approved for triple-negative BC.^{79,80} CV toxic effects associated with ICIs have been reviewed elsewhere^{14,79} and are summarized in Table 1. ICI-related myocarditis is a rare (incidence <1%) but potentially fatal complication, with a case-fatality rate up to 25% to 50%.¹⁴ ICIs are also associated with pericarditis, vasculitis, and Takotsubo cardiomyopathy. They may accelerate atherosclerosis through activation of inflammatory pathways, but this clinical impact remains unclear.¹⁴

Radiation Therapy

Breast and chest radiation reduces the risk of local recurrence, but cardiac exposure is associated with an increased risk of CV events.^{81,82} However, modern radiation therapy techniques for BC have advanced substantially to minimize cardiac exposure, as discussed later.

Cardioprotective Strategies in Patients With BC

Most research on cardioprotection has focused on anthracycline cardiotoxic effects. The cardioprotective strategies described below have varying strengths of evidence to support their use, and this must be kept in mind when determining their use in clinical practice.

Anthracycline Administration and Dexrazoxane

Because anthracyclines exhibit dose-dependent cardiotoxic effects, limiting their dose can reduce cardiotoxic effects, especially in patients with high CV risk. Continuous infusion and liposomal doxorubicin may decrease cardiotoxic effects without affecting cancer outcomes.^{83,84} Liposomal doxorubicin, encapsulated within a phospholipid bilayer, limits diffusion into healthy tissues while maintaining tissue cell penetrance.

Dexrazoxane mitigates anthracycline cardiotoxic effects by preventing oxygen-derived free radical generation and inhibiting the interaction of anthracyclines with topoisomerase II β . Most studies on dexrazoxane as a cardioprotective agent come from childhood cancer research, where it was administered concurrently with anthracyclines from the initiation of treatment and was shown to be effective for doses of 250 mg/m² or greater.⁸⁵ In a meta-analysis, dexrazoxane significantly reduced clinical HF risk and cardiac events without compromising cancer outcomes.⁸⁶ In advanced BC, dexrazoxane was still cardioprotective when initiated after a cumulative anthracycline dose of 300 mg/m².⁸⁷ As a result, dexrazoxane is currently approved in the US for patients receiving at least 300 mg/m² of anthracyclines, but use in BC treatment is relatively infrequent. This is in part because oncologists are uncertain what total anthracycline dose will be administered when treating without curative intent (because treatment may be changed to a nonanthracycline at the time of disease progression). Liposomal doxorubicin is often used in this setting for this reason.

The 2022 ESC cardio-oncology guidelines provide a class IIa, level of evidence B recommendation for the use of liposomal doxorubicin or dexrazoxane in patients at high risk for cardiotoxic effects and a class IIb, level of evidence C recommendation in those with preexisting systolic dysfunction who require anthracycline therapy.⁵⁹ To our knowledge, no studies have directly compared the

cardioprotective effects of dexrazoxane and liposomal doxorubicin. In practice, dexrazoxane is preferred for many pediatric patients and for adults who have already received a high cumulative dose of standard anthracycline.

Neurohormonal Blockade and Statins

Studies evaluating prophylactic statin or neurohormonal blockade have shown mixed results⁸⁸⁻¹⁰¹ (Table 2). The STOP-CA (Statins to Prevent the Cardiotoxicity of Anthracyclines) trial,⁸⁸ which enrolled patients with lymphoma receiving higher-dose anthracyclines (median dose, 300 mg/m²), showed that prophylactic statins may ameliorate cardiotoxic effects in patients receiving anthracyclines, but results from studies in patients with BC with lower anthracycline exposure have been neutral.^{89,90} Meta-analyses suggest potential benefit, but significant heterogeneity limits clear conclusions.

Similarly, based on conflicting data, the 2022 ESC cardio-oncology guidelines give a class IIa, level of evidence B recommendation for prophylactic neurohormonal blockade in patients receiving anthracyclines or anti-ERBB2 therapies and at high or very high risk for developing HF. For patients who are receiving other targeted cancer therapies that may cause HF and who are at high or very high risk for HF, the 2022 ESC cardio-oncology guidelines give a class IIa, level of evidence C recommendation for prophylactic neurohormonal blockade. The guidelines provide a class IIa, level of evidence B recommendation for statins in patients at high or very high risk of cardiotoxic effects.⁵⁹ However, the 2022 American College of Cardiology/American Heart Association/Heart Failure Society of America guidelines conclude that prophylactic neurohormonal blockade in high-risk patients is of uncertain benefit, giving it a class IIb, level of evidence B-R recommendation.¹⁰² Regardless, all patients should be screened for CV risk factors or CVD and have treatment with statins or neurohormonal blockade initiated if clinically indicated.

Radiation Delivery Strategies to Limit Cardiac Exposure in BC

Advances in radiation therapy have promoted cardiac sparing to decrease cardiotoxic effects from radiation.^{81,103-105} Radiation during deep inspiratory breath hold, which physically moves the heart from the treatment field, has arguably had the greatest impact on reducing cardiac exposure.^{106,107} Recent guideline updates have expanded the use of partial breast radiation in patients with early-stage BC who are older than 40 years, as this technique reduces the size of the radiation field, further reducing cardiac risk.¹⁰⁸ Proton therapy, unlike the more commonly used photon therapy, offers significant potential to reduce cardiac exposure due to its physical properties that limit tissue penetration. Dosimetric studies consistently show a 5- to 10-fold reduction in dose to the heart, left anterior descending artery, and ventricle with proton therapy.^{109,110} The recently completed Pragmatic Randomised Trial of Proton vs Photon Therapy for Patients With Nonmetastatic Breast Cancer (RadComp) trial will provide data on the effectiveness of proton vs photon therapy in reducing major CV events.¹¹¹ However, proton therapy is not widely available and is limited by higher costs.

Exercise

There is growing appreciation of the importance of exercise, including aerobic and resistance training, during BC treatment to improve long-term outcomes,¹¹²⁻¹¹⁶ but its effects on LV function preservation are mixed. Although the impact of exercise on the

Table 2. Major Randomized Clinical Trials of Neurohormonal Blockade and Statin Therapy for Cardioprotection Against Anthracycline- and Trastuzumab-Induced Cardiotoxic Effects in Patients With Breast Cancer (BC)

Trial	Disease	Study design	No.	Intervention	Follow-up	Primary outcome	Result
PRADA II ⁹²	Early-stage BC	Placebo-controlled, double-blind, multicenter trial of anthracycline and cyclophosphamide	138	1:1 Sacubitril-valsartan:placebo	18 mo	Change in CMR LVEF from baseline	No significant difference in primary outcome with ARNI
SARAH ⁹³	BC	Placebo-controlled, double-blind, single-center trial of anthracyclines plus evidence of cardiac injury (troponin I level >99th percentile) after anthracycline infusion	114	1:1 Sacubitril-valsartan:placebo	24 wk	≥15% Reduction in GLS from baseline	Primary outcome occurred less with ARNI (7.1%) vs placebo (25.0%)
PRADA ⁹⁴	Early-stage BC	Placebo-controlled, double-blind, single-center trial of anthracyclines with or without trastuzumab	120	1:1:1:1 Candesartan + metoprolol succinate:candesartan:metoprolol succinate:placebo	2 y	Change in CMR LVEF from baseline	No significant difference with candesartan or metoprolol succinate vs placebo
SAFE ⁹⁵	Early-stage breast CA	Placebo-controlled, double-blind, multicenter trial of anthracyclines with or without trastuzumab	174	1:1:1:1 Bisoprolol + ramipril:bisoprolol:ramipril:placebo	12 mo	Decrease in LVEF or GLS ≥10% from baseline	Both end points significantly reduced with ramipril and bisoprolol
CECCY ⁹⁶	ERBB2-negative, early-stage BC	Placebo-controlled, double-blind, single-center trial of anthracyclines	192	1:1 Carvedilol:placebo	6 mo	Decrease in LVEF ≥10% from baseline	No significant difference with carvedilol
PROACT ⁹⁷	BC, NHL	Open-label, multicenter trial of anthracyclines ≥300 mg/m ²	111	1:1 Enalapril:standard care	During or 1 mo after anthracycline treatment	cTnT ≥14 ng/L	No significant difference with enalapril
MANTICORE ⁹⁸	ERBB2-positive, early-stage BC	Placebo-controlled, double-blind, multicenter trial of trastuzumab with or without anthracyclines	94	1:1:1 Perindopril:bisoprolol:placebo	End of trastuzumab	Change in LVEDVI from baseline to end of trastuzumab therapy	No significant difference with perindopril or bisoprolol vs placebo
Boekhout et al ⁹⁹	ERBB2-positive, early-stage BC	Placebo-controlled, double-blind, multicenter trial of anthracyclines and trastuzumab	210	1:1 Candesartan:placebo	2 y	Decrease in LVEF >15% from baseline or an LVEF <45%	No significant change with candesartan
Guglin et al ¹⁰⁰	ERBB2-positive BC	Placebo-controlled, double-blind, multicenter trial of trastuzumab with or without anthracyclines	468	1:1:1 Carvedilol:lisinopril:placebo	2 y	Decrease in LVEF >10% from baseline or >5% from baseline to LVEF <50%	No benefit in overall population, but decreased cardiotoxic effects with carvedilol and lisinopril in patients treated with anthracyclines but not trastuzumab alone
Cardiac CARE ⁹¹	BC, NHL	Open-label, blinded-end point, multicenter trial of anthracyclines	175	1:1 Carvedilol + candesartan; standard care if high risk by troponin I level	6 mo	Change in LVEF at 6 mo	No benefit in patients at high risk (elevated on-treatment cardiac troponin I concentrations)
Nabati et al ¹⁰¹	BC (100%) ^a	Placebo-controlled, single-blind, single-center trial of anthracyclines with or without trastuzumab	77	1:1 Rosuvastatin:placebo	6 mo	Change in LVEF and GLS from baseline after completion of chemotherapy	No significant change in LVEF in statin group; significant decrease in LVEF in placebo group; no significant difference in GLS
PREVENT ⁸⁹	BC (85%) ^a	Placebo-controlled, double-blind, multicenter trial of anthracyclines with or without trastuzumab (median doxorubicin dose 240 mg/m ²)	279	1:1 Atorvastatin:placebo	24 mo	Change in CMR LVEF from baseline	No difference with atorvastatin
SPARE-HF ⁹⁰	BC (65%) ^a	Placebo-controlled, double-blind, multicenter trial of anthracyclines	112	1:1 Atorvastatin:placebo	2.5 mo	Change in CMR LVEF from baseline	No difference with atorvastatin
STOP-CA ⁸⁸	Lymphoma	Placebo-controlled, double-blind, multicenter trial of anthracyclines (median doxorubicin dose 300 mg/m ²)	300	1:1 Atorvastatin:placebo	12 mo	Decrease in LVEF ≥10% from baseline to LVEF <55%	Lower incidence of primary outcome in statin vs placebo group (9% vs 22%; P = .002)

Abbreviations: ARNI, angiotensin receptor–neprilysin inhibitor; Cardiac CARE, High-Sensitivity Cardiac Troponin

I-Guided Combination Angiotensin Receptor Blockade and Beta Blocker Therapy to Prevent Cardiac Toxicity

in Cancer Patients Receiving Anthracycline Chemotherapy; CECCY, Carvedilol Effect in Preventing

Chemotherapy-Induced Cardiotoxicity; CMR, cardiac magnetic resonance imaging; cTnT, cardiac troponin T;

GLS, global longitudinal strain; LVEF, left ventricular ejection fraction; LVEDVI, left ventricular end-diastolic

volume index; MANTICORE, Multidisciplinary Approach to Novel Therapies in Cardio-Oncology Research;

NHL, non-Hodgkin lymphoma; PRADA, Prevention of Cardiac Dysfunction During Adjuvant Breast Cancer

Therapy; PROACT, Preventing Cardiac Damage in Patients Treated for Breast Cancer and Lymphoma;

SARAH, Sacubitril-Valsartan for the Prevention of Anthracycline Cardiotoxicity in Patients With Elevated hs-cTnI

Concentrations During Chemotherapy; SPARE-HF, Statins for the Primary Prevention of Heart Failure in Patients

With Cancer Receiving Anthracycline-Based Chemotherapy; STOP-CA, Statins to Prevent the Cardiotoxicity

of Anthracyclines.

^a Percentage indicates the percentage of individuals with BC enrolled in the trial.

prevention of cardiotoxic effects in BC is uncertain, the 2022 ESC cardio-oncology guidelines recommend at least 150 min/wk of moderate exercise for patients undergoing cancer therapy.⁵⁹

Exercise is also important to improve outcomes after cancer treatment.¹¹⁷ In survivors of BC, cardiorespiratory fitness declines significantly following cancer therapy to predicted levels of sedentary, healthy women 20 to 30 years older than the patient's chronological age.¹¹⁸ This reduction may improve but typically does not return to pretreatment levels. Increased physical activity is associated with a graded reduction in the incidence of CV events in women with nonmetastatic BC¹¹⁹ and higher global longitudinal strain (GLS) in survivors of BC.¹²⁰ Whether the improved disease-free and overall survival with structured exercise in patients with colon cancer will be reproduced in patients with BC remains to be seen.¹²¹

CVD Prevention and Treatment for Patients With a History of BC

CV assessment before, during, and after BC therapy is essential (Figure 2). Recommendations regarding the type and frequency of CV testing differ between established guidelines,^{15,59,102} as summarized in Table 3. General guiding principles are discussed below, with specific guidelines referenced as appropriate.

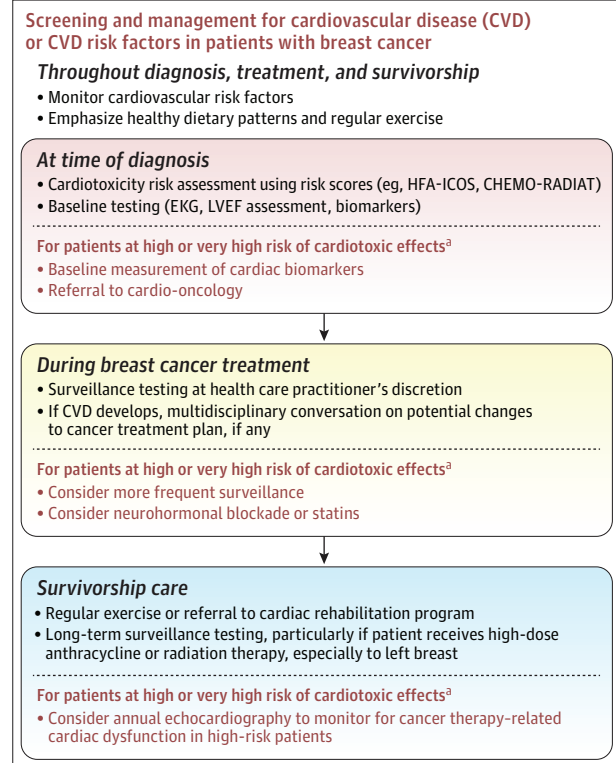
Assessment Prior to Initiation of Cancer Treatment

Patients should be screened for traditional CV risk factors and pre-existing CVD, and CV health should be optimized prior to the initiation of potentially cardiotoxic treatments.^{15,59,102,122} A baseline electrocardiogram and assessment of cardiac function (via transthoracic echocardiography preferably with GLS, cardiac magnetic resonance imaging [MRI], or multigated acquisition nuclear imaging) may be useful, particularly in selected patients with known CVD or those receiving therapies with potential for CTRCD.^{15,59,102}

Several risk assessment tools have been developed to predict cardiotoxic effects.¹⁶ The European Heart Failure Association–International Cardio-Oncology Society score assigns patients into low-, moderate-, high-, and very high-risk categories for developing CTRCD and has been validated in patients with BC treated with anthracyclines and trastuzumab.^{17,123} The CHEMO-RADIAT risk score, designed to predict long-term major adverse CV events in survivors of BC, also demonstrates strong predictive accuracy at 7 years following diagnosis.¹²⁴ These risk scores allow for personalized management and informed decision-making regarding choice of cancer therapy by balancing cancer treatment with CV risk and deciding on preventive and monitoring strategies, including which patients would be appropriate for referral to cardio-oncology.

The role of baseline cardiac biomarker measurement to aid in risk stratification is controversial, and guideline recommendations differ (Table 3). Small, prospective studies (n = 78–251) demonstrated elevations in high-sensitivity troponin level during anthracycline therapy¹²⁵ and increased risk of cardiotoxic effects induced by trastuzumab¹²⁶ or trastuzumab plus anthracycline^{127,128} in those with elevations of troponin I during treatment. In a recent study, neurohormonal blockade was not cardioprotective in high-risk individuals, defined as having elevations in troponin I level during anthracycline therapy.⁹¹ Additionally, the degree of LVEF decline in high-risk patients was similar to that in low-risk patients, who had

Figure 2. A Life-Course Approach to Preventing and Managing Cardiovascular Risk in Patients With Breast Cancer



CHEMO-RADIAT indicates Congestive Heart Failure, Hypertension, Elderly, Myocardial Infarction/Peripheral Artery Occlusive Disease, Obesity, Renal Failure, Abnormal Lipid Profile, Diabetes Mellitus, Irradiation of the Left Breast, Anthracycline Dose, and Transient Ischemic Attack/Stroke; CVD, cardiovascular disease; EKG, electrocardiogram; HFA-ICOS, Heart Failure Association–International Cardio-Oncology Society; LVEF, left ventricular ejection fraction.

^a2022 European Society of Cardiology cardio-oncology guidelines.

lower troponin I concentrations, calling into question the utility of troponin I level in risk stratification.⁹¹ N-terminal pro-B-type natriuretic peptide levels have not been associated with cardiotoxic effects.^{127,128} To our knowledge, there have been no studies on biomarkers for risk stratification of delayed (ie, years after completion of treatment) clinical presentation of CTRCD.

Monitoring for Cardiotoxic Effects During Cancer Treatment

Monitoring for cardiotoxic effects during cancer treatment should be individualized based on preexisting CVD, risk for de novo or worsening CVD, and treatment received.^{15,59,102}

For individuals considered to be at high or very high risk for developing CTRCD, preventive therapy with neurohormonal blockade or statins can be considered,⁵⁹ and recent studies have attempted to ascertain the benefits of GLS- or biomarker-guided initiation. The SUCCOUR (Strain Surveillance of Chemotherapy for Improving Cardiovascular Outcomes) trial found no difference in 3-year echocardiographic LVEF between GLS- and LVEF-guided strategies, as well as no significant difference in the incidence of CTRCD (11 vs 6 patients; *P* = .16). In contrast, the SUCCOUR MRI trial randomized patients who developed a decline in GLS, without corresponding changes in LVEF, to neurohormonal blockade or usual

Table 3. Cardiac Surveillance in Asymptomatic Patients Receiving Select Cardiotoxic Breast Cancer Therapies

	Risk for cardiotoxic effects by therapy (class of recommendation, LOE)									
Testing per guideline	Anthracyclines ^a			Trastuzumab ^a			Pembrolizumab ^a			Ribociclib
	Low	Moderate	High	Low	Moderate	High	Low	Moderate	High	
LVEF ^b										
ESC ⁵⁹	Baseline, 12 mo after treatment (I, B); cycle 4 (IIb, C)	Baseline, 12 mo after treatment (I, B); cycle 4 (IIa, C)	Baseline, cycles 2, 4, and 6, 3 mo, and 12 mo after treatment (I, C)	Baseline, every 3 mo, and 12 mo after treatment (I, B)	Baseline, every 3 mo, and 12 mo after treatment (I, B)	Baseline, every 3 mo, then 3 mo and 12 mo after treatment (I, B)	Baseline (IIb, C)	NA	Baseline (I, B)	NA
ASCO ¹⁵	Routine surveillance echocardiogram may be offered for asymptomatic patients considered at increased risk, with frequency to be determined by health care practitioner and individualized for the patient; criteria for increased risk include the following: (1) treatment with any of the following: high-dose anthracycline (eg, doxorubicin ≥250 mg/m ² , epirubicin ≥600 mg/m ²), high-dose radiotherapy (≥30 Gy) with heart in the treatment field, or lower-dose anthracycline (eg, doxorubicin <250 mg/m ² , epirubicin <600 mg/m ²) with lower-dose RT (<30 Gy) with heart in the treatment field; (2) treatment with lower-dose anthracycline or trastuzumab alone and presence of any of the following: ≥2 CV RFs (including smoking, hypertension, type 2 diabetes, hyperlipidemia, obesity during or after completion of therapy), aged ≥60 y at cancer treatment, borderline low LVEF (50%-55%), history of MI, at least moderate valvular heart disease before or during treatment; (3) treatment with lower-dose anthracycline followed by trastuzumab (sequential therapy), echocardiogram may be performed 6-12 mo after completion of therapy in patients considered to be at increased risk; may use routine echocardiographic surveillance in patients with metastatic breast cancer continuing to receive trastuzumab indefinitely									
ACC/AHA/HFSA ¹⁰²	Baseline echocardiogram and serial monitoring for patients receiving anthracyclines and anti-ERBB2 agents, frequency at the discretion of the health care practitioner (IIa)									
Biomarkers										
ESC ⁵⁹	Baseline (IIa, C), cycles 2, 4, and 6, and 3 mo after treatment (IIb, C)	Baseline, cycles 2, 4, and 6, and 3 mo after treatment (IIa, C)	Baseline, cycles 1-6, and 3 mo and 12 mo after treatment (I, B)	Baseline, every 3 mo, and 12 mo after treatment (IIb, C)	Baseline, every 3 mo, and 12 mo after treatment (IIb, C)	Baseline (I, C), every 3 mo, and 12 mo after treatment (IIa, C)	Baseline (I, B), cTn: cycles 2-4 and then every 3 cycles until completion of therapy (IIa, B), NP: every 6-12 mo (IIb, C) ^c	NA	Baseline (I, B), cTn: cycles 2-4 and then every 3 cycles until completion of therapy (IIa, B), NP: every 6-12 mo (I, C) ^c	NA
ASCO ¹⁵	The use of standard biomarkers in asymptomatic patients during and after completing cancer-directed therapy remains investigational; further studies are needed									
ACC/AHA/HFSA ¹⁰²	Patients being considered for potentially cardiotoxic therapies, serial cardiac troponin measurements might be reasonable (IIb)									
EKG										
ESC ⁵⁹	Baseline (I, C)	Baseline (I, C)	Baseline (I, C)	Baseline (I, C)	Baseline (I, C)	Baseline (I, C)	Baseline (I, B), cycles 2-4 and then every 3 cycles until completion of treatment (IIa, B), every 6-12 mo (IIb, C) ^c	NA	Baseline (I, B), cycles 2-4 and then every 3 cycles until completion of treatment (IIa, B), every 6-12 mo (I, C) ^c	Baseline, day 14, beginning of second cycle (ie, day 28) and with any dose increase
ASCO ¹⁵	No specific recommendation									
ACC/AHA/HFSA ¹⁰²	No specific recommendation									

Abbreviations: ACC, American College of Cardiology; AHA, American Heart Association; ASCO, American Society of Clinical Oncology; cTn, cardiac troponin; CV, cardiovascular; EKG, electrocardiogram; ESC, European Society of Cardiology; HFSA, Heart Failure Society of America; LOE, level of evidence; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NA, not applicable; NP, natriuretic peptide; RFs, risk factors; RT, radiation therapy.

^a Consider using validated risk score to assess risk and guide frequency of

surveillance.

^b Two- or three-dimensional transthoracic echocardiogram with global longitudinal strain preferred; cardiac magnetic resonance imaging or multigated acquisition nuclear imaging if poor echocardiographic windows.

^c In patients who require long-term (>12 -month) immune checkpoint inhibitor treatment.

care and showed better preservation of LVEF on MRI at 12 months with neurohormonal blockade.^{129,130} Several studies have failed to show a definitive benefit of biomarker-guided initiation of neurohormonal blockade.^{91,131} Most trials have enrolled patients irrespective of CTRCD risk and may have diluted potential benefit in high-risk patients.

If individuals develop signs or symptoms of cardiac dysfunction, diagnostic testing to assess LV systolic function and serum biomarkers should be performed.^{15,59} There is uncertain benefit to adjusting cancer treatment based on GLS measurements, and it is not

currently advised. Decisions on continuation or discontinuation of cancer therapy based on changes in LVEF should be individualized with multidisciplinary discussion, balancing the extent of malignancy with the risk of further cardiac decompensation.^{15,59,102}

Permissive Cardiotoxicity

Given worse cancer outcomes when cancer therapies are prematurely discontinued or interrupted,¹³² there is growing interest in *permissive cardiotoxicity*. Primarily studied with trastuzumab, it suggests the continued use of potentially cardiotoxic therapies

in patients with asymptomatic, mild cardiotoxic effects (LVEF 40%-49%).¹³³ The SAFE-HEaRt trial enrolled patients with stage I through IV ERBB2-positive BC who had asymptomatic, mildly depressed LVEF (40%-49%). All had neurohormonal blockade initiated prior to planned anti-ERBB2 therapy. The primary outcome of completing planned anti-ERBB2 therapy was achieved in 90% of individuals without significant worsening of LV dysfunction or symptomatic HF.¹³⁴ These findings extended into long-term follow-up.⁶³ The SCHOLAR I (Safety of Continuing Chemotherapy in Overt Left Ventricular Dysfunction Using Antibodies to Human Epidermal Growth Factor Receptor-2) trial⁶⁴ enrolled patients receiving trastuzumab with evidence of myocardial injury, defined as LVEF of 40% to 54% or with a decline of at least 15% from baseline. Patients began treatment with angiotensin-converting enzyme inhibitors and/or β -blockers and were followed up closely. Among enrolled patients, 90% completed trastuzumab therapy and 10% experienced dose-limiting cardiotoxic effects (LVEF <40%).⁶⁴

Both the SAFE-HEaRt and SCHOLAR I trials were small ($n = 30$ and 20 , respectively), open-label, single-arm studies, thus limiting the strength of their conclusions. The larger SCHOLAR-2 trial is currently ongoing (ClinicalTrials.gov identifier: [NCT04680442](https://clinicaltrials.gov/ct2/show/study/NCT04680442)). Additionally, while permissive cardiotoxicity has potential, it requires close monitoring and involvement of a multidisciplinary cardio-oncology team.

Long-Term Surveillance and Cancer Survivorship

It is projected that by 2030, there will be nearly 5 million survivors of BC in the US,² an increasing proportion of whom are living with metastatic disease, all needing multidisciplinary survivorship care.^{2,117,135} These patients are at long-term CV risk either from adjuvant therapy after completion of treatment (ie, adjuvant ET for 5-10 years)⁶⁶ or CTRCD, which can occur decades after completion of treatment.^{44,81} While long-term posttreatment surveillance for adverse CV effects is recommended for patients with BC, there is an interest in identifying patients in whom deescalation of cardiac monitoring may be appropriate. Those living with metastatic cancer may receive anti-ERBB2 therapies for many years (sometimes decades), in which case deescalation of surveillance may be considered to minimize cost and burden to patients. If there is no evidence of cardiac compromise on every-3-month LVEF monitoring during the first 1 to 2 years of anti-ERBB2 therapy, the risk of developing clinically significant echocardiographic abnormalities over the subsequent years is thought to be low, especially if the patient has never received an anthracycline as part of their treatment. Oncologists may thus reduce the frequency of subsequent echocardiograms after discussion with the patient.

For long-term survivors of BC, preexisting CV comorbidities can be a greater threat to health than the cancer itself.⁵ The ESC guidelines recommend annual CV risk factor assessment and aggressive risk factor modification in all survivors of BC (Figure 2).⁵⁹ While undergoing treatment for cancer, a decrease in adherence to noncancer therapies, such as antihypertensive agents or lipid-lowering therapies, has been observed, possibly exacerbating the CV risk from anticancer

therapies.⁶⁸ Radiation to the left breast or mediastinum may be considered a risk enhancer,¹³⁶ and factoring in radiation dose to the heart may help identify higher-risk patients in need of CV optimization by a cardiologist.⁸² Breast artery calcifications, incidentally identified either on screening mammograms or on pretreatment computed tomographic scans, are associated with future atherosclerotic CVD risk, suggesting a role for more aggressive primary prevention.^{137,138}

Exercise plays a key role in cancer survivorship, as discussed earlier. Emerging calls for the development of cardiac rehabilitation programs for survivors of BC emphasize the importance of early intervention and specific adaptation to maximize the physical and psychosocial health of women.¹³⁹ Beyond improving cardiorespiratory fitness, programs have also been shown to reduce fatigue and improve quality of life in survivors of BC.¹¹⁷

Gaps in Knowledge and Future Directions

The rapid evolution of BC treatments, particularly immunotherapy and targeted therapies, necessitates concurrent research on the risk for and prevention of CVD with novel therapies. The additive risk imposed by sequential cardiotoxic therapies also deserves attention. There are limited data on the optimal timing and frequency of imaging and biomarker measurement for CVD monitoring, and the ideal balance between early CVD detection and overtesting is debated. Risk scores have been proposed, but further precision to identify patients who may benefit from more or less frequent surveillance is needed. Additional studies in patients with metastatic disease receiving prolonged cardiotoxic therapies, and what is appropriate surveillance for this group of patients, are needed. Most individuals enrolled in studies evaluating neurohormonal blockade or statin therapy for prevention of cardiotoxic effects had few to no risk factors for CVD. Additional studies on higher-risk patients are needed. Newer medications, such as sodium-glucose cotransporter 2 inhibitors, glucagon-like peptide-1 receptor agonists, and other incretin-based therapies, have expanded indications in CVD and require further investigation in cardio-oncology. Several trials on methods to prevent CV toxic effects after cancer therapy and on withdrawal of neurohormonal blockade in those patients who have recovered LV systolic function have completed enrollment or are still enrolling.^{78,140-142} Whether structured exercise will improve BC disease-free survival is unknown. Given the close monitoring needed for permissive cardiotoxicity, if it is used more broadly, issues regarding access to multidisciplinary cardio-oncology services must be addressed.

Conclusions

Advancements in the treatment and survival of BC have made CVD prevention and management increasingly important across the life course of patients with BC. As cancer therapeutics evolve, so must research on ways to screen for, prevent, and manage CVD and CV risk factors during and after BC therapy.

ARTICLE INFORMATION

Accepted for Publication: March 29, 2026.

Published Online: July 29, 2026.
doi:[10.1001/jamacardio.2026.2546](https://doi.org/10.1001/jamacardio.2026.2546)

Author Affiliations: Department of Medicine, Division of Cardiology, University of Texas Southwestern, Dallas (Mauricio); Department of

Cardiology, Division of Internal Medicine, University of Texas MD Anderson Cancer Center, Houston (Deswal); Department of Oncology, Mayo Clinic, Rochester, Minnesota (Ruddy); Department of Medicine, Division of Cardiology, Northwestern University Feinberg School of Medicine, Chicago, Illinois (Akhter); Cardiovascular Health for Cancer Survivors Program, Department of Pediatrics, Division of Genetics and Genomics, Boston Children's Hospital and Dana-Farber Cancer Institute, Boston, Massachusetts (Chen); Division of Cardiovascular Medicine, Department of Internal Medicine, University of California Davis, Sacramento (Ebong); Department of Cardiovascular Disease, Mayo Clinic, Rochester, Minnesota (Aggarwal); Division of Cardiology, Department of Medicine, Massachusetts General Hospital, Harvard Medical School, Boston (Harrington); Department of Breast Medical Oncology, University of Texas MD Anderson Cancer Center, Houston, Texas (Barcenas); Department of Breast Radiation Oncology, University of Texas MD Anderson Cancer Center, Houston (Mitchell); Department of Cardiovascular Medicine, Heart Vascular Thoracic Institute, Cleveland Clinic, Cleveland Ohio (Cho); Department of Internal Medicine, Division of Cardiology, Brigham and Women's Hospital, Boston, Massachusetts (Nohria).

Author Contributions: Drs Mauricio and Deswal had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Mauricio, Deswal, Akhter, Ebong, Aggarwal, Barcenas, Mitchell, Cho, Nohria. **Acquisition, analysis, or interpretation of data:** Ruddy, Chen, Harrington.

Drafting of the manuscript: Mauricio, Deswal, Akhter, Chen, Ebong, Aggarwal, Harrington, Barcenas, Mitchell.

Critical review of the manuscript for important intellectual content: Mauricio, Deswal, Ruddy, Akhter, Chen, Ebong, Aggarwal, Harrington, Barcenas, Cho, Nohria.

Statistical analysis: Mauricio.

Administrative, technical, or material support: Ebong.

Supervision: Mauricio, Deswal, Nohria.

Conflict of Interest Disclosures: Dr Deswal reported personal fees from Bayer and the Ting Tsung and Wei Fong Chao Distinguished Chair from MD Anderson Cancer Center outside the submitted work. Dr Ruddy reported that her spouse is coinventor of a technology licensed by Mayo Clinic to Alivacor related to the application of artificial intelligence to the electrocardiogram outside the submitted work. Dr Akhter reported personal fees from Zoll and speaking for Cleerly (no fees received) outside the submitted work. Dr Nohria reported personal fees from AstraZeneca, Takeda Oncology, Medical Learning Institute, and American College of Cardiology and grants from Bristol Myers Squibb outside the submitted work. No other disclosures were reported.

REFERENCES

1. Siegel RL, Giaquinto AN, Jemal A. Cancer statistics, 2024. *CA Cancer J Clin*. 2024;74(1):12-49.
2. Miller KD, Nogueira L, Devasia T, et al. Cancer treatment and survivorship statistics, 2022. *CA Cancer J Clin*. 2022;72(5):409-436.
3. National Cancer Institute. Cancer stat facts: female breast cancer, 2025. Accessed February 19,

2025. <https://seer.cancer.gov/statfacts/html/breast.html>

4. Heer E, Harper A, Escandor N, Sung H, McCormack V, Fidler-Benaoudia MM. Global burden and trends in premenopausal and postmenopausal breast cancer: a population-based study. *Lancet Glob Health*. 2020;8(8):e1027-e1037. doi:10.1016/S2214-109X(20)30215-1
5. Mehta LS, Watson KE, Barac A, et al; American Heart Association Cardiovascular Disease in Women and Special Populations Committee of the Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Cardiovascular disease and breast cancer: where these entities intersect: a scientific statement from the American Heart Association. *Circulation*. 2018;137(8):e30-e66. doi:10.1161/CIR.0000000000000556
6. Shah AN, Gradishar WJ. Adjuvant anthracyclines in breast cancer: what is their role? *Oncologist*. 2018;23(10):1153-1161. doi:10.1634/theoncologist.2017-0672
7. Cardinale D, Colombo A, Bacchiani G, et al. Early detection of anthracycline cardiotoxicity and improvement with heart failure therapy. *Circulation*. 2015;131(22):1981-1988. doi:10.1161/CIRCULATIONAHA.114.013777
8. Slamon DJ, Leyland-Jones B, Shak S, et al. Use of chemotherapy plus a monoclonal antibody against HER2 for metastatic breast cancer that overexpresses HER2. *N Engl J Med*. 2001;344(11):783-792. doi:10.1056/NEJM200103153441101
9. de Azambuja E, Ponde N, Procter M, et al. A pooled analysis of the cardiac events in the trastuzumab adjuvant trials. *Breast Cancer Res Treat*. 2020;179(1):161-171. doi:10.1007/s10549-019-05453-z
10. Okwuosa TM, Morgans A, Rhee JW, et al; American Heart Association Cardio-Oncology Subcommittee of the Council on Clinical Cardiology and the Council on Genomic and Precision Medicine; Council on Arteriosclerosis, Thrombosis and Vascular Biology; and Council on Cardiovascular Radiology and Intervention. Impact of hormonal therapies for treatment of hormone-dependent cancers (breast and prostate) on the cardiovascular system: effects and modifications: a scientific statement from the American Heart Association. *Circ Genom Precis Med*. 2021;14(3):e000082. doi:10.1161/HCG.0000000000000082
11. Thomas NS, Scalzo RL, Wellberg EA. Diabetes mellitus in breast cancer survivors: metabolic effects of endocrine therapy. *Nat Rev Endocrinol*. 2024;20(1):16-26. doi:10.1038/s41574-023-00899-0
12. Honigberg MC, Zekavat SM, Aragam K, et al. Association of premature natural and surgical menopause with incident cardiovascular disease. *JAMA*. 2019;322(24):2411-2421. doi:10.1001/jama.2019.19191
13. Arnett DK, Blumenthal RS, Albert MA, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2019;140(11):e563-e595. doi:10.1161/CIR.0000000000000677
14. Palaskas NL, Ali HJ, Koutroumpakis E, Ganatra S, Deswal A. Cardiovascular toxicity of immune therapies for cancer. *BMJ*. 2024;385:e075859. doi:10.1136/bmj-2023-075859
15. Armenian SH, Lacchetti C, Barac A, et al. Prevention and monitoring of cardiac dysfunction in survivors of adult cancers: American Society of Clinical Oncology Clinical Practice Guideline. *J Clin Oncol*. 2017;35(8):893-911. doi:10.1200/JCO.2016.70.5400
16. Lyon AR, Dent S, Stanway S, et al. Baseline cardiovascular risk assessment in cancer patients scheduled to receive cardiotoxic cancer therapies: a position statement and new risk assessment tools from the Cardio-Oncology Study Group of the Heart Failure Association of the European Society of Cardiology in collaboration with the International Cardio-Oncology Society. *Eur J Heart Fail*. 2020;22(11):1945-1960. doi:10.1002/ehf.1920
17. Ali A, Koutroumpakis E, Song J, et al. Risk stratification for trastuzumab-induced cardiac dysfunction and potential implications for surveillance. *JACC CardioOncol*. 2025;7(3):203-215. doi:10.1016/j.jacc.2024.12.007
18. Duan J, Tao J, Zhai M, et al. Anticancer drugs-related QTc prolongation, torsade de pointes and sudden death: current evidence and future research perspectives. *Oncotarget*. 2018;9(39):25738-25749. doi:10.18632/oncotarget.25008
19. Frickhofen N, Beck FJ, Jung B, Fuhr HG, Andrasch H, Sigmund M. Capecitabine can induce acute coronary syndrome similar to 5-fluorouracil. *Ann Oncol*. 2002;13(5):797-801. doi:10.1093/annonc/mdf035
20. Kwakman JJ, Simkens LH, Mol L, Kok WE, Koopman M, Punt CJ. Incidence of capecitabine-related cardiotoxicity in different treatment schedules of metastatic colorectal cancer: a retrospective analysis of the CAIRO studies of the Dutch Colorectal Cancer Group. *Eur J Cancer*. 2017;76:93-99. doi:10.1016/j.ejca.2017.02.009
21. Slamon D, Eiermann W, Robert N, et al; Breast Cancer International Research Group. Adjuvant trastuzumab in HER2-positive breast cancer. *N Engl J Med*. 2011;365(14):1273-1283. doi:10.1056/NEJMoa0910383
22. de Azambuja E, Agostinetto E, Procter M, et al; APHINITY Steering Committee and Investigators. Cardiac safety of dual anti-HER2 blockade with pertuzumab plus trastuzumab in early HER2-positive breast cancer in the APHINITY trial. *ESMO Open*. 2023;8(1):100772. doi:10.1016/j.esmoop.2022.100772
23. Dent SF, Morse A, Burnette S, Guha A, Moore H. Cardiovascular toxicity of novel HER2-targeted therapies in the treatment of breast cancer. *Curr Oncol Rep*. 2021;23(11):128. doi:10.1007/s11912-021-01114-x
24. Davies C, Godwin J, Gray R, et al; Early Breast Cancer Trialists' Collaborative Group (EBCTCG). Relevance of breast cancer hormone receptors and other factors to the efficacy of adjuvant tamoxifen: patient-level meta-analysis of randomised trials. *Lancet*. 2011;378(9793):771-784. doi:10.1016/S0140-6736(11)60993-8
25. Fisher B, Costantino JP, Wickerham DL, et al. Tamoxifen for prevention of breast cancer: report of the National Surgical Adjuvant Breast and Bowel Project P-1 Study. *J Natl Cancer Inst*. 1998;90(18):1371-1388. doi:10.1093/jnci/90.18.1371
26. Cuzick J, Forbes J, Edwards R, et al; IBIS investigators. First results from the International Breast Cancer Intervention Study (IBIS-I): a randomised prevention trial. *Lancet*. 2002;360(9336):817-824. doi:10.1016/S0140-6736(02)09962-2
27. Khosrow-Khavar F, Filion KB, Al-Qurashi S, et al. Cardiotoxicity of aromatase inhibitors and tamoxifen in postmenopausal women with breast cancer: a systematic review and meta-analysis of

- randomized controlled trials. *Ann Oncol*. 2017;28(3):487-496. doi:10.1093/annonc/mdw673
28. Boszkiewicz K, Piwowar A, Petryszyn P. Aromatase inhibitors and risk of metabolic and cardiovascular adverse effects in breast cancer patients—a systematic review and meta-analysis. *J Clin Med*. 2022;11(11):11. doi:10.3390/jcm11113133
29. Sehl ME, Ganz PA. Potential mechanisms of age acceleration caused by estrogen deprivation: do endocrine therapies carry the same risks? *J Natl Cancer Inst Cancer Spectr*. 2018;2(3):pk035. doi:10.1093/jncics/pky035
30. Francis PA, Regan MM, Fleming GF, et al; SOFT Investigators; International Breast Cancer Study Group. Adjuvant ovarian suppression in premenopausal breast cancer. *N Engl J Med*. 2015;372(5):436-446. doi:10.1056/NEJMoa1412379
31. Rocca WA, Gazzuola-Rocca L, Smith CY, et al. Accelerated accumulation of multimorbidity after bilateral oophorectomy: a population-based cohort study. *Mayo Clin Proc*. 2016;91(11):1577-1589. doi:10.1016/j.mayocp.2016.08.002
32. Grey AB, Stapleton JP, Evans MC, Reid IR. The effect of the anti-estrogen tamoxifen on cardiovascular risk factors in normal postmenopausal women. *J Clin Endocrinol Metab*. 1995;80(11):3191-3195.
33. Buzdar A, Howell A, Cuzick J, et al; Arimidex, Tamoxifen, Alone or in Combination Trialists' Group. Comprehensive side-effect profile of anastrozole and tamoxifen as adjuvant treatment for early-stage breast cancer: long-term safety analysis of the ATAC trial. *Lancet Oncol*. 2006;7(8):633-643. doi:10.1016/S1470-2045(06)70767-7
34. Bundred NJ. The effects of aromatase inhibitors on lipids and thrombosis. *Br J Cancer*. 2005;93(suppl 1):S23-S27. doi:10.1038/sj.bjc.6602692
35. Markopoulos C, Chrissochou M, Michailidou A, et al. Effect of exemestane on the lipidemic profile of post-menopausal operable breast cancer patients following 5-7 years of adjuvant tamoxifen: preliminary results of the ATENA substudy. *Anticancer Drugs*. 2005;16(8):879-883. doi:10.1097/01.cad.0000173478.12981.e1
36. Hozumi Y, Suemasu K, Takei H, et al. The effect of exemestane, anastrozole, and tamoxifen on lipid profiles in Japanese postmenopausal early breast cancer patients: final results of National Surgical Adjuvant Study BC 04, the TEAM Japan sub-study. *Ann Oncol*. 2011;22(8):1777-1782. doi:10.1093/annonc/mdq707
37. Younus M, Kissner M, Reich L, Wallis N. Putting the cardiovascular safety of aromatase inhibitors in patients with early breast cancer into perspective: a systematic review of the literature. *Drug Saf*. 2011;34(12):1125-1149. doi:10.2165/11594170-000000000-00000
38. Price MA, Alvarado BE, Rosendaal NTA, Câmara SMA, Pirkle CM, Velez MP. Early and surgical menopause associated with higher Framingham Risk Scores for cardiovascular disease in the Canadian Longitudinal Study on Aging. *Menopause*. 2021;28(5):484-490. doi:10.1097/GME.0000000000001729
39. Watson NW, Shatzel JJ, Al-Samkari H. Cyclin-dependent kinase 4/6 inhibitor-associated thromboembolism: a critical evaluation of the current evidence. *J Thromb Haemost*. 2023;21(4):758-770. doi:10.1016/j.jtha.2022.12.001
40. Fradley MG, Nguyen NHK, Madnick D, et al. Adverse cardiovascular events associated with cyclin-dependent kinase 4/6 inhibitors in patients with metastatic breast cancer. *J Am Heart Assoc*. 2023;12(12):e029361. doi:10.1161/JAHA.123.029361
41. Murthy RK, Loi S, Okines A, et al. Tucatinib, trastuzumab, and capecitabine for HER2-positive metastatic breast cancer. *N Engl J Med*. 2020;382(7):597-609. doi:10.1056/NEJMoa1914609
42. Aapro M, Cardoso F, Curigliano G, et al. Current challenges and unmet needs in treating patients with human epidermal growth factor receptor 2-positive advanced breast cancer. *Breast*. 2022;66:145-156. doi:10.1016/j.breast.2022.07.011
43. André F, Ciruelos E, Rubovszky G, et al; SOLAR-1 Study Group. Apalissib for *PIK3CA*-mutated, hormone receptor-positive advanced breast cancer. *N Engl J Med*. 2019;380(20):1929-1940. doi:10.1056/NEJMoa1813904
44. Camilli M, Cipolla CM, Dent S, Minotti G, Cardinale DM. Anthracycline cardiotoxicity in adult cancer patients: JACC: CardioOncology state-of-the-art review. *JACC CardioOncol*. 2024;6(5):655-677. doi:10.1016/j.jaccao.2024.07.016
45. Rasmussen-Torvik LJ, Shay CM, Abramson JG, et al. Ideal cardiovascular health is inversely associated with incident cancer: the Atherosclerosis Risk in Communities study. *Circulation*. 2013;127(12):1270-1275. doi:10.1161/CIRCULATIONAHA.112.001183
46. Zhao Y, Song Y, Li X, Guo A. Association of Life's Essential 8 cardiovascular health with breast cancer incidence and mortality according to genetic susceptibility of breast cancer: a prospective cohort study. *Breast Cancer Res*. 2024;26(1):121. doi:10.1186/s13058-024-01877-8
47. Foraker RE, Abdel-Rasoul M, Kuller LH, et al. Cardiovascular health and incident cardiovascular disease and cancer: the Women's Health Initiative. *Am J Prev Med*. 2016;50(2):236-240. doi:10.1016/j.amepre.2015.07.039
48. Ogunmoroti O, Allen NB, Cushman M, et al. Association between Life's Simple 7 and noncardiovascular disease: the Multi-Ethnic Study of Atherosclerosis. *J Am Heart Assoc*. 2016;5(10):5. doi:10.1161/JAHA.116.003954
49. Newman AAC, Dalman JM, Moore KJ. Cardiovascular disease and cancer: a dangerous liaison. *Arterioscler Thromb Vasc Biol*. 2025;45(3):359-371. doi:10.1161/ATVBAHA.124.319863
50. Jain V, Rifai MA, Brinzevich D, et al. Association of premature atherosclerotic cardiovascular disease with higher risk of cancer: a behavioral risk factor surveillance system study. *Eur J Prev Cardiol*. 2022;29(3):493-501. doi:10.1093/eurjpc/zwab084
51. Koelwyn GJ, Newman AAC, Afonso MS, et al. Myocardial infarction accelerates breast cancer via innate immune reprogramming. *Nat Med*. 2020;26(9):1452-1458. doi:10.1038/s41591-020-0964-7
52. Meijers WC, Maglione M, Bakker SJL, et al. Heart failure stimulates tumor growth by circulating factors. *Circulation*. 2018;138(7):678-691. doi:10.1161/CIRCULATIONAHA.117.030816
53. Tani T, Oikawa M, Misaka T, Ishida T, Takeishi Y. Heart failure post-myocardial infarction promotes mammary tumor growth through the NGF-TRKA pathway. *JACC CardioOncol*. 2023;6(1):55-66. doi:10.1016/j.jaccao.2023.10.002
54. Yuan Y, Mei Z, Qu Z, et al. Exosomes secreted from cardiomyocytes suppress the sensitivity of tumor ferroptosis in ischemic heart failure. *Signal Transduct Target Ther*. 2023;8(1):121. doi:10.1038/s41392-023-01336-4
55. Callier T, Rotem I, Shaihov-Teper O, et al. Small extracellular vesicles from infarcted and failing heart accelerate tumor growth. *Circulation*. 2024;149(22):1729-1748. doi:10.1161/CIRCULATIONAHA.123.066911
56. Koutrompakis E, Agrawal N, Palaskas NL, et al. Myocardial dysfunction in patients with cancer. *Heart Fail Clin*. 2022;18(3):361-374. doi:10.1016/j.hfc.2022.02.011
57. Bloom MW, Vo JB, Rodgers JE, et al. Cardio-oncology and heart failure: a scientific statement from the Heart Failure Society of America. *J Card Fail*. 2025;31(2):415-455. doi:10.1016/j.cardfail.2024.08.045
58. Tarantino P, Curigliano G, Parsons HA, et al. Aiming at a tailored cure for *ERBB2*-positive metastatic breast cancer: a review. *JAMA Oncol*. 2022;8(4):629-635. doi:10.1001/jamaoncol.2021.6597
59. Lyon AR, López-Fernández T, Couch LS, et al; ESC Scientific Document Group. 2022 ESC guidelines on cardio-oncology developed in collaboration with the European Hematology Association (EHA), the European Society for Therapeutic Radiology and Oncology (ESTRO) and the International Cardio-Oncology Society (IC-OS). *Eur Heart J*. 2022;43(41):4229-4361. doi:10.1093/eurheartj/ehac244
60. Dempsey N, Rosenthal A, Dabas N, Kropotova Y, Lippman M, Bishopric NH. Trastuzumab-induced cardiotoxicity: a review of clinical risk factors, pharmacologic prevention, and cardiotoxicity of other HER2-directed therapies. *Breast Cancer Res Treat*. 2021;188(1):21-36. doi:10.1007/s10549-021-06280-x
61. Ewer MS, Vooletich MT, Durand JB, et al. Reversibility of trastuzumab-related cardiotoxicity: new insights based on clinical course and response to medical treatment. *J Clin Oncol*. 2005;23(31):7820-7826. doi:10.1200/JCO.2005.13.300
62. Barac A, Blaes A, Lynce F. Lessons from primary cardiac prevention trials during trastuzumab therapy: end of one size fits all. *J Am Coll Cardiol*. 2019;73(22):2869-2871. doi:10.1016/j.jacc.2019.04.006
63. Khoury K, Lynce F, Barac A, et al. Long-term follow-up assessment of cardiac safety in SAFE-HEaRT, a clinical trial evaluating the use of HER2-targeted therapies in patients with breast cancer and compromised heart function. *Breast Cancer Res Treat*. 2021;185(3):863-868. doi:10.1007/s10549-020-06053-y
64. Leong DP, Cosman T, Alhussein MM, et al. Safety of continuing trastuzumab despite mild cardiotoxicity: a phase I trial. *JACC CardioOncol*. 2019;1(1):1-10. doi:10.1016/j.jaccao.2019.06.004
65. Zhou S, Cirne F, Chow J, et al. Three-year outcomes following permissive cardiotoxicity in patients on trastuzumab. *Oncologist*. 2023;28(9):e712-e722. doi:10.1093/oncolo/oyad086
66. Gradishar WJ, Moran MS, Abraham J, et al. Breast cancer, version 3.2024. NCCN Clinical Practice Guidelines in Oncology. *J Natl Compr Canc Netw*. 2024;22(5):331-357. doi:10.6004/jnccn.2024.0035
67. Rillamas-Sun E, Kwan ML, Iribarren C, et al. Development of cardiometabolic risk factors following endocrine therapy in women with breast cancer. *Breast Cancer Res Treat*. 2023;201(1):117-126. doi:10.1007/s10549-023-06997-x
68. Zullig LL, Sung AD, Khouri MG, et al. Cardiometabolic comorbidities in cancer survivors: JACC: CardioOncology state-of-the-art review. *JACC CardioOncol*. 2022;4(2):149-165. doi:10.1016/j.jaccao.2022.03.005

69. Khosrow-Khavar F, Filion KB, Bouganim N, Suissa S, Azoulay L. Aromatase inhibitors and the risk of cardiovascular outcomes in women with breast cancer: a population-based cohort study. *Circulation*. 2020;141(7):549-559. doi:10.1161/CIRCULATIONAHA.119.044750
70. Sund M, Garcia-Arribay M, Garmo H, et al. Aromatase inhibitors use and risk for cardiovascular disease in breast cancer patients: a population-based cohort study. *Breast*. 2021;59:157-164. doi:10.1016/j.breast.2021.07.004
71. van Ommen-Nijhof A, Jacobse JN, Steggink LC, et al. Adjuvant aromatase inhibitor therapy and early markers for cardiovascular disease in breast cancer survivors. *Breast Cancer Res Treat*. 2022;196(3):591-602. doi:10.1007/s10549-022-06714-0
72. Kamaraju S, Shi Y, Smith E, Nattinger AB, Laud P, Neuner J. Are aromatase inhibitors associated with higher myocardial infarction risk in breast cancer patients? a Medicare population-based study. *Clin Cardiol*. 2019;42(1):93-100. doi:10.1002/clc.23114
73. Francis PA, Fleming GF, Láng I, et al; SOFT Investigators and the International Breast Cancer Study Group (a Division of ETOP IBCSG Partners Foundation). Adjuvant endocrine therapy in premenopausal breast cancer: 12-year results from SOFT. *J Clin Oncol*. 2023;41(7):1370-1375. doi:10.1200/JCO.22.01065
74. Baek SY, Noh WC, Ahn SH, et al. Adding ovarian suppression to tamoxifen for premenopausal women with hormone receptor-positive breast cancer after chemotherapy: an 8-year follow-up of the ASTRRA trial. *J Clin Oncol*. 2023;41(31):4864-4871. doi:10.1200/JCO.23.00557
75. Francis PA, Pagani O, Fleming GF, et al; SOFT and TEXT Investigators and the International Breast Cancer Study Group. Tailoring adjuvant endocrine therapy for premenopausal breast cancer. *N Engl J Med*. 2018;379(2):122-137. doi:10.1056/NEJMoA1803164
76. Parker WH, Feskanich D, Broder MS, et al. Long-term mortality associated with oophorectomy compared with ovarian conservation in the Nurses' Health Study. *Obstet Gynecol*. 2013;121(4):709-716. doi:10.1097/AOG.0b013e3182864350
77. Muka T, Oliver-Williams C, Kunutsor S, et al. Association of age at onset of menopause and time since onset of menopause with cardiovascular outcomes, intermediate vascular traits, and all-cause mortality: a systematic review and meta-analysis. *JAMA Cardiol*. 2016;1(7):767-776. doi:10.1001/jamacardio.2016.2415
78. Thomas A, O'Connell NS, Douglas E, et al. Cardiovascular impact of near complete estrogen deprivation in premenopausal women with breast cancer: the CROWN study. *Am Heart J*. 2024;267:33-43. doi:10.1016/j.ahj.2023.10.007
79. Schmid P, Cortes J, Pusztai L, et al; KEYNOTE-522 Investigators. Pembrolizumab for early triple-negative breast cancer. *N Engl J Med*. 2020;382(9):810-821. doi:10.1056/NEJMoA1910549
80. Schmid P, Cortes J, Dent R, et al; KEYNOTE-522 Investigators. Overall survival with pembrolizumab in early-stage triple-negative breast cancer. *N Engl J Med*. 2024;391(21):1981-1991. doi:10.1056/NEJMoA2409932
81. Darby SC, Ewertz M, McGale P, et al. Risk of ischemic heart disease in women after radiotherapy for breast cancer. *N Engl J Med*. 2013;368(11):987-998. doi:10.1056/NEJMoA1209825
82. van den Bogaard VA, Ta BD, van der Schaaf A, et al. Validation and modification of a prediction model for acute cardiac events in patients with breast cancer treated with radiotherapy based on three-dimensional dose distributions to cardiac substructures. *J Clin Oncol*. 2017;35(11):1171-1178. doi:10.1200/JCO.2016.69.8480
83. Smith LA, Cornelius VR, Plummer CJ, et al. Cardiotoxicity of anthracycline agents for the treatment of cancer: systematic review and meta-analysis of randomised controlled trials. *BMC Cancer*. 2010;10:337. doi:10.1186/1471-2407-10-337
84. Xing M, Yan F, Yu S, Shen P. Efficacy and cardiotoxicity of liposomal doxorubicin-based chemotherapy in advanced breast cancer: a meta-analysis of ten randomized controlled trials. *PLoS One*. 2015;10(7):e0133569. doi:10.1371/journal.pone.0133569
85. Chow EJ, Aggarwal S, Doody DR, et al. Dexrazoxane and long-term heart function in survivors of childhood cancer. *J Clin Oncol*. 2023;41(12):2248-2257. doi:10.1200/JCO.22.02423
86. Macedo AVS, Hajjar LA, Lyon AR, et al. Efficacy of dexrazoxane in preventing anthracycline cardiotoxicity in breast cancer. *JACC CardioOncol*. 2019;1(1):68-79. doi:10.1016/j.jacc.2019.08.003
87. Swain SM, Whaley FS, Gerber MC, Ewer MS, Bianchini JR, Gams RA. Delayed administration of dexrazoxane provides cardioprotection for patients with advanced breast cancer treated with doxorubicin-containing therapy. *J Clin Oncol*. 1997;15(4):1333-1340. doi:10.1200/JCO.1997.15.4.1333
88. Neilan TG, Quinaglia T, Onoue T, et al. Atorvastatin for anthracycline-associated cardiac dysfunction: the STOP-CA randomized clinical trial. *JAMA*. 2023;330(6):528-536. doi:10.1001/jama.2023.11887
89. Jandub W, D'Agostino R Jr, Crotts T, et al. Statins and left ventricular ejection fraction following doxorubicin treatment. *NEJM Evid*. 2022;1(9):1. doi:10.1056/EVIDoA2200097
90. Thavendiranathan P, Houbos C, Marwick TH, et al. Statins to prevent early cardiac dysfunction in cancer patients at increased cardiotoxicity risk receiving anthracyclines. *Eur Heart J Cardiovasc Pharmacother*. 2023;9(6):515-525. doi:10.1093/ehjcvp/pvad031
91. Henriksen PA, Hall P, MacPherson IR, et al. Multicenter, prospective, randomized controlled trial of high-sensitivity cardiac troponin I-guided combination angiotensin receptor blockade and beta-blocker therapy to prevent anthracycline cardiotoxicity: the Cardiac CARE trial. *Circulation*. 2023;148(21):1680-1690. doi:10.1161/CIRCULATIONAHA.123.064274
92. Omland T, Heck SL, Holte E, et al. Sacubitril/valsartan and prevention of cardiac dysfunction during adjuvant breast cancer therapy: the PRADA II randomized clinical trial. *Circulation*. 2025;152(16):1136-1145. doi:10.1161/CIRCULATIONAHA.125.076616
93. Bonatto MG, Avila MS, Ayub Ferreira SM, et al. Sacubitril-valsartan for the prevention of anthracycline cardiotoxicity in patients with elevated cardiac troponin I concentration during chemotherapy: a double-blind randomized placebo-controlled clinical trial: the SARAH trial. *Circulation*. 2025;152(25):1742-1755. doi:10.1161/CIRCULATIONAHA.125.073322
94. Heck SL, Mecinaj A, Ree AH, et al. Prevention of cardiac dysfunction during adjuvant breast cancer therapy (PRADA): extended follow-up of a 2x2 factorial, randomized, placebo-controlled, double-blind clinical trial of candesartan and metoprolol. *Circulation*. 2021;143(25):2431-2440. doi:10.1161/CIRCULATIONAHA.121.054698
95. Livi L, Barletta G, Martella F, et al. Cardioprotective strategy for patients with nonmetastatic breast cancer who are receiving an anthracycline-based chemotherapy: a randomized clinical trial. *JAMA Oncol*. 2021;7(10):1544-1549. doi:10.1001/jamaoncol.2021.3395
96. Avila MS, Ayub-Ferreira SM, de Barros Wanderley MR Jr, et al. Carvedilol for prevention of chemotherapy-related cardiotoxicity: the CECY trial. *J Am Coll Cardiol*. 2018;71(20):2281-2290. doi:10.1016/j.jacc.2018.02.049
97. Austin D, Maier RH, Akhter N, et al. Preventing cardiac damage in patients treated for breast cancer and lymphoma: the PROACT clinical trial. *JACC CardioOncol*. 2024;6(5):684-696. doi:10.1016/j.jacc.2024.07.010
98. Pituskin E, Mackey JR, Koshman S, et al. Multidisciplinary approach to novel therapies in cardio-oncology research (MANTICORE 101-Breast): a randomized trial for the prevention of trastuzumab-associated cardiotoxicity. *J Clin Oncol*. 2017;35(8):870-877. doi:10.1200/JCO.2016.68.7830
99. Boekhout AH, Gietema JA, Milojkovic Kerklaan B, et al. Angiotensin II-receptor inhibition with candesartan to prevent trastuzumab-related cardiotoxic effects in patients with early breast cancer: a randomized clinical trial. *JAMA Oncol*. 2016;2(8):1030-1037. doi:10.1001/jamaoncol.2016.1726
100. Guglin M, Krischer J, Tamura R, et al. Randomized trial of lisinopril versus carvedilol to prevent trastuzumab cardiotoxicity in patients with breast cancer. *J Am Coll Cardiol*. 2019;73(22):2859-2868. doi:10.1016/j.jacc.2019.03.495
101. Nabati M, Janbabai G, Esmailian J, Yazdani J. Effect of rosuvastatin in preventing chemotherapy-induced cardiotoxicity in women with breast cancer: a randomized, single-blind, placebo-controlled trial. *J Cardiovasc Pharmacol Ther*. 2019;24(3):233-241. doi:10.1177/1074248418821721
102. Heidenreich PA, Bozkurt B, Aguilar D, et al; ACC/AHA Joint Committee Members. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032. doi:10.1161/CIR.0000000000001063
103. Pierce LJ, Butler JB, Martel MK, et al. Postmastectomy radiotherapy of the chest wall: dosimetric comparison of common techniques. *Int J Radiat Oncol Biol Phys*. 2002;52(5):1220-1230. doi:10.1016/S0360-3016(01)02760-2
104. Lymberis SC, deWynngaert JK, Parhar P, et al. Prospective assessment of optimal individual position (prone versus supine) for breast radiotherapy: volumetric and dosimetric correlations in 100 patients. *Int J Radiat Oncol Biol Phys*. 2012;84(4):902-909. doi:10.1016/j.ijrobp.2012.01.040
105. Hamming VC, Visser C, Batin E, et al. Evaluation of a 3D surface imaging system for deep inspiration breath-hold patient positioning and intra-fraction monitoring. *Radiat Oncol*. 2019;14(1):125. doi:10.1186/s13014-019-1329-6
106. Lu HM, Cash E, Chen MH, et al. Reduction of cardiac volume in left-breast treatment fields by respiratory maneuvers: a CT study. *Int J Radiat Oncol Biol Phys*. 2000;47(4):895-904. doi:10.1016/S0360-3016(00)00512-5
107. Chen MH, Chuang ML, Bornstein BA, Gelman R, Harris JR, Manning WJ. Impact of

- respiratory maneuvers on cardiac volume within left-breast radiation portals. *Circulation*. 1997;96(10):3269-3272. doi:10.1161/01.CIR.96.10.3269
108. Shaitelman SF, Anderson BM, Arthur DW, et al. Partial breast irradiation for patients with early-stage invasive breast cancer or ductal carcinoma in situ: an ASTRO Clinical Practice Guideline. *Pract Radiat Oncol*. 2024;14(2):112-132. doi:10.1016/j.prro.2023.11.001
109. Loap P, Tkatchenko N, Goudjil F, et al. Cardiac substructure exposure in breast radiotherapy: a comparison between intensity modulated proton therapy and volumetric modulated arc therapy. *Acta Oncol*. 2021;60(8):1038-1044. doi:10.1080/0284186X.2021.1907860
110. Bradley JA, Dagan R, Ho MW, et al. Initial report of a prospective dosimetric and clinical feasibility trial demonstrates the potential of protons to increase the therapeutic ratio in breast cancer compared with photons. *Int J Radiat Oncol Biol Phys*. 2016;95(1):411-421. doi:10.1016/j.ijrobp.2015.09.018
111. Bekelman JE, Lu H, Pugh S, et al; RadComp (Radiotherapy Comparative Effectiveness Consortium). Pragmatic randomised clinical trial of proton versus photon therapy for patients with non-metastatic breast cancer: the Radiotherapy Comparative Effectiveness (RadComp) Consortium trial protocol. *BMJ Open*. 2019;9(10):e025556. doi:10.1136/bmjopen-2018-025556
112. Ligibel JA, Bohlke K, May AM, et al. Exercise, diet, and weight management during cancer treatment: ASCO guideline. *J Clin Oncol*. 2022;40(22):2491-2507. doi:10.1200/JCO.22.00687
113. Díaz-Balboa E, Peña-Gil C, Rodríguez-Romero B, et al. Exercise-based cardio-oncology rehabilitation for cardiotoxicity prevention during breast cancer chemotherapy: the ONCORE randomized controlled trial. *Prog Cardiovasc Dis*. 2024;85:74-81. doi:10.1016/j.pcad.2024.02.002
114. van der Schoot GGF, Ormel HL, Westerink NL, et al. Optimal timing of a physical exercise intervention to improve cardiorespiratory fitness: during or after chemotherapy. *JACC CardioOncol*. 2022;4(4):491-503. doi:10.1016/j.jacc.2022.07.006
115. Foulkes SJ, Howden EJ, Haykowsky MJ, et al. Exercise for the prevention of anthracycline-induced functional disability and cardiac dysfunction: the BREXIT study. *Circulation*. 2023;147(7):532-545. doi:10.1161/CIRCULATIONAHA.122.062814
116. Linhares BG, Linhares DG, Boppre G, Zacca R. New insights into cardioprotection in breast cancer patients undergoing physical exercise during chemotherapy: a systematic review and meta-analysis. *Curr Probl Cardiol*. 2024;49(10):102743. doi:10.1016/j.cpcardiol.2024.102743
117. Gilchrist SC, Barac A, Ades PA, et al; American Heart Association Exercise, Cardiac Rehabilitation, and Secondary Prevention Committee of the Council on Clinical Cardiology; Council on Cardiovascular and Stroke Nursing; and Council on Peripheral Vascular Disease. Cardio-oncology rehabilitation to manage cardiovascular outcomes in cancer patients and survivors: a scientific statement from the American Heart Association. *Circulation*. 2019;139(21):e997-e1012. doi:10.1161/CIR.0000000000000679
118. Jones LW, Courneya KS, Mackey JR, et al. Cardiopulmonary function and age-related decline across the breast cancer survivorship continuum. *J Clin Oncol*. 2012;30(20):2530-2537. doi:10.1200/JCO.2011.39.9014
119. Jones LW, Habel LA, Weltzien E, et al. Exercise and risk of cardiovascular events in women with nonmetastatic breast cancer. *J Clin Oncol*. 2016;34(23):2743-2749. doi:10.1200/JCO.2015.65.6603
120. Naaktgeboren WR, Groen WG, Jacobse JN, et al. Physical activity and cardiac function in long-term breast cancer survivors: a cross-sectional study. *JACC CardioOncol*. 2022;4(2):183-191. doi:10.1016/j.jacc.2022.02.007
121. Courneya KS, Vardy JL, O'Callaghan CJ, et al; CHALLENGE Investigators. Structured exercise after adjuvant chemotherapy for colon cancer. *N Engl J Med*. 2025;393(1):13-25. doi:10.1056/NEJMoa2502760
122. Rao VU, Deswal A, Lenihan D, et al. Quality-of-care measures for cardio-oncology: an IC-OS and ACC Cardio-Oncology Leadership Council perspective. *JACC CardioOncol*. 2025;7(3):191-202. doi:10.1016/j.jacc.2024.11.003
123. Rivero-Santana B, Saldaña-García J, Caro-Codón J, et al. Anthracycline-induced cardiovascular toxicity: validation of the Heart Failure Association and International Cardio-Oncology Society risk score. *Eur Heart J*. 2025;46(3):273-284. doi:10.1093/eurheartj/ehae496
124. Kim DY, Park MS, Youn JC, et al. Development and validation of a risk score model for predicting the cardiovascular outcomes after breast cancer therapy: the CHERO-RADIAT score. *J Am Heart Assoc*. 2021;10(16):e021931. doi:10.1161/JAHA.121.021931
125. Tzolos E, Adamson PD, Hall PS, et al. Dynamic Changes in high-sensitivity cardiac troponin I in response to anthracycline-based chemotherapy. *Clin Oncol (R Coll Radiol)*. 2020;32(5):292-297. doi:10.1016/j.clon.2019.11.008
126. Cardinale D, Colombo A, Torrisi R, et al. Trastuzumab-induced cardiotoxicity: clinical and prognostic implications of troponin I evaluation. *J Clin Oncol*. 2010;28(25):3910-3916. doi:10.1200/JCO.2009.27.3615
127. Ky B, Putt M, Sawaya H, et al. Early increases in multiple biomarkers predict subsequent cardiotoxicity in patients with breast cancer treated with doxorubicin, taxanes, and trastuzumab. *J Am Coll Cardiol*. 2014;63(8):809-816. doi:10.1016/j.jacc.2013.10.061
128. Sawaya H, Sebag IA, Plana JC, et al. Assessment of echocardiography and biomarkers for the extended prediction of cardiotoxicity in patients treated with anthracyclines, taxanes, and trastuzumab. *Circ Cardiovasc Imaging*. 2012;5(5):596-603. doi:10.1161/CIRCIMAGING.112.973321
129. Thavendiranathan P, Negishi T, Somerset E, et al; SUCCOUR Investigators. Strain-guided management of potentially cardiotoxic cancer therapy. *J Am Coll Cardiol*. 2021;77(4):392-401. doi:10.1016/j.jacc.2020.11.020
130. Marwick TH, Dewar E, Nolan M, et al. Strain surveillance during chemotherapy to improve cardiovascular outcomes: the SUCCOUR-MRI trial. *Eur Heart J*. 2024;45(41):4414-4424. doi:10.1093/eurheartj/ehae574
131. Cardinale D, Ciceri F, Latini R, et al; ICOS-ONE Study Investigators. Anthracycline-induced cardiotoxicity: a multicenter randomised trial comparing two strategies for guiding prevention with enalapril: the International CardioOncology Society-one trial. *Eur J Cancer*. 2018;94:126-137. doi:10.1016/j.ejca.2018.02.005
132. Copeland-Halperin RS, Al-Sadawi M, Patil S, et al. Early trastuzumab interruption and recurrence-free survival in ERBB2-positive breast cancer. *JAMA Oncol*. 2020;6(12):1971-1972. doi:10.1001/jamaoncol.2020.4749
133. Porter C, Azam TU, Mohananeey D, et al. Permissive cardiotoxicity: the clinical crucible of cardio-oncology. *JACC CardioOncol*. 2022;4(3):302-312. doi:10.1016/j.jacc.2022.07.005
134. Lynce F, Barac A, Geng X, et al. Prospective evaluation of the cardiac safety of HER2-targeted therapies in patients with HER2-positive breast cancer and compromised heart function: the SAFE-HEaRt study. *Breast Cancer Res Treat*. 2019;175(3):595-603. doi:10.1007/s10549-019-05191-2
135. Runowicz CD, Leach CR, Henry NL, et al. American Cancer Society/American Society of Clinical Oncology breast cancer survivorship care guideline. *CA Cancer J Clin*. 2016;66(1):43-73.
136. Carlson LE, Watt GP, Tonorez ES, et al; WE CARE Study Collaborative Group. Coronary artery disease in young women after radiation therapy for breast cancer: the WE CARE Study. *JACC CardioOncol*. 2021;3(3):381-392. doi:10.1016/j.jacc.2021.07.008
137. Gal R, van Velzen SGM, Hoening MJ, et al. Identification of risk of cardiovascular disease by automatic quantification of coronary artery calcifications on radiotherapy planning CT scans in patients with breast cancer. *JAMA Oncol*. 2021;7(7):1024-1032. doi:10.1001/jamaoncol.2021.1144
138. Allen TS, Bui QM, Petersen GM, et al. Automated breast arterial calcification score is associated with cardiovascular outcomes and mortality. *JACC Adv*. 2024;3(11):101283. doi:10.1016/j.jacadv.2024.101283
139. Chen MH, Epstein SF. Tailored to a woman's heart: gender cardio-oncology across the lifespan. *Curr Cardiol Rep*. 2023;25(11):1461-1474. doi:10.1007/s11886-023-01967-7
140. Mecinaj A, Gulati G, Heck SL, et al. Rationale and design of the Prevention of Cardiac Dysfunction During Adjuvant Breast Cancer Therapy (PRADA II) trial: a randomized, placebo-controlled, multicenter trial. *Cardiooncology*. 2021;7(1):33. doi:10.1186/s40959-021-00115-w
141. Shibutani Y, Imaoka T, Suzuki A, Tajiri K. Withdrawal of cardioprotective therapy in patients with recovered cancer therapy-related cardiac dysfunction. *JACC CardioOncol*. 2025;7(2):189. doi:10.1016/j.jacc.2024.11.010
142. Dowsing B, Dehbi HM, Chung R, et al. HER-SAFE study design: an open-label, randomised controlled trial to investigate the safety of withdrawal of pharmacological treatment for recovered HER2-targeted therapy-related cardiac dysfunction. *BMJ Open*. 2025;15(2):e091917. doi:10.1136/bmjopen-2024-091917